



1.12

Transformations of Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

| Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: For $g(x) = af(b(x-h)) + k$, input transformations occur inside and output transformations occur outside.

2. Write one precise sentence explaining this principle: Horizontal scale is reciprocal: b produces factor $1/|b|$.

3. Write one precise sentence explaining this principle: A negative a reflects across the x-axis; a negative b reflects across the y-axis.

4. Write one precise sentence explaining this principle: Track a point $(u, f(u))$ by solving $b(x-h) = u$.

Core

5. Describe $g(x) = -3f(2(x-4)) + 5$.

6. If $(6, -2)$ lies on f , find the corresponding point on $g(x) = 2f(x + 1) - 3$.

7. Why does replacing x by $x-5$ shift right rather than left?

8. State what the formula $g(x) = af(b(x - h)) + k$ tells you and name any required restriction.

9. State what the formula $(u, f(u)) \mapsto (h + \frac{u}{b}, af(u) + k)$ tells you and name any required restriction.

10. Create a second representation (table, graph, formula, or context) for one central relationship in Transformations of Functions. State its domain and justify one key feature.

Medium

11. Describe the transformations in $g(x) = f(x - 4) + 3$.

12. Describe the transformations in $g(x) = -2f(x + 1)$.

13. Describe the transformations in $g(x) = f(3x)$.

14. Describe the transformations in $g(x) = f(-x) + 5$.

15. Describe the transformations in $g(x) = \frac{1}{2}f(x/4) - 2$.

Hard

16. Point $(-2, 5)$ lies on f . Find its image under $g(x) = -3f(2(x - 1)) + 4$.

17. Write g in terms of f if its graph is f reflected over y -axis, stretched horizontally by 3, and shifted down 2.

18. Find the vertex and range of $g(x) = -2|x + 3| + 7$.

19. Compare $f(2x - 6)$ and $f(2(x - 6))$.

Difficult

20. If $g(x) = af(b(x - h)) + k$ maps point $(4, 2)$ to $(1, -5)$, give equations relating a, b, h, k .

21. Find a transformation of $f(x) = x^2$ whose graph has vertex $(3, -4)$ and passes through $(1, 8)$.

22. Determine the domain of $g(x) = \sqrt{2x - 6} + 1$ using transformation and algebra.

Challenge / HOT

23. Prove the point mapping formula for $g(x) = af(b(x - h)) + k$.

24. Can two different transformation descriptions produce the same function? Give an example.

AP-Style Multiple Choice

25. $f(x - 3)$ shifts
 (A) left 3
 (B) right 3
 (C) up 3
 (D) down 3
26. $f(2x)$ is a horizontal
 (A) stretch 2
 (B) compression $1/2$

- (C) shift 2
(D) reflection
27. Negative outside f reflects across
(A) x-axis
(B) y-axis
(C) $y=x$
(D) origin only
28. Point (2,5) on f maps under $g(x) = f(x + 4) - 1$ to
(A) (6,4)
(B) (-2,4)
(C) (2,8)
(D) (-2,6)
29. Vertex of $3(x + 1)^2 - 2$ is
(A) (1,-2)
(B) (-1,-2)
(C) (-1,2)
(D) (3,-2)
30. Inside horizontal scale uses
(A) same factor
(B) reciprocal factor
(C) no factor
(D) squared factor

AP-Style Free Response

31. Let f have domain $[-4, 6]$, range $[-2, 8]$, and point (2, 5). Define $g(x) = -2f(3(x - 1)) + 4$. Find (a) domain, (b) range, (c) image of the point.

32. A transformed square-root function has endpoint (5,-3), decreases as x increases, and passes through (9,-7). Construct a formula and state domain/range.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.